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FlowBridge: A Zero-Cost Generalized Framework for Cross-Platform Email-to-Messaging Automation Using Browser Automation in Containerized Virtual Display Environments

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April 2026

Source code: github.com/trivenigk/flowbridge-job-forwarder

ABSTRACT

We present **FlowBridge**, a generalized, zero-cost framework for building automated email-to-messaging pipelines that monitor email inboxes, extract structured content, deduplicate across temporal windows, and deliver formatted alerts to messaging platform groups using browser automation within containerized virtual display environments. FlowBridge introduces several novel subsystems applicable to any email-to-messaging use case: (1) **DedupGuard**, a multi-signal deduplication engine combining sender fingerprinting, content similarity analysis, and unique identifier matching at the pipeline boundary between email ingestion and message dispatch; (2) **ClipCast**, a clipboard-based Unicode injection method that bypasses WebDriver Basic Multilingual Plane limitations for emoji-rich formatted messages; (3) **SessionHeal**, a self-healing browser session management mechanism for unattended containerized operation; (4) **SheetConfig**, a spreadsheet-as-configuration pattern enabling non-technical users to manage pipeline behavior including message routing, recipient groups, and processing rules without code changes; and (5) **BatchForge**, a message aggregation engine that combines multiple content items into a single formatted transmission, reducing recipient notification fatigue. We demonstrate FlowBridge through a production deployment forwarding job postings from Gmail to WhatsApp groups, achieving 100% delivery accuracy, zero duplicate messages, and zero recurring infrastructure cost. Prior art analysis confirms that while individual components exist in isolation, the integrated FlowBridge pipeline architecture represents a novel contribution not covered by any existing patent or publication.

Keywords: FlowBridge, email automation, messaging pipeline, browser automation, containerized virtual display, deduplication, zero-cost infrastructure, cross-platform messaging, spreadsheet configuration

1. Introduction

1.1 Problem Statement

Organizations and communities face a persistent challenge: critical information arrives via email (notifications, alerts, reports, postings) while team communication happens in messaging platforms (WhatsApp, Telegram, Slack, Discord). Bridging this gap manually is tedious, error-prone, and inconsistent.

This problem manifests across diverse domains:

Domain	Email Source	Messaging Target	Content
Job communities	Recruiter emails, job boards	WhatsApp/Telegram groups	Job postings
Sales teams	Lead notifications, CRM alerts	Slack channels	New leads, deal updates
DevOps	Monitoring alerts, CI/CD notifications	Discord/Teams channels	Incident alerts
Real estate	Property listings, MLS updates	WhatsApp groups	New listings
Academic research	Paper alerts, conference notifications	Lab group chats	Relevant papers
E-commerce	Price drop alerts, stock notifications	Customer groups	Deal alerts

1.2 Existing Approaches and Limitations

API-based integrations (Zapier, Make.com, n8n): Require official messaging platform APIs. WhatsApp Business API costs $0.005-0.08/message and requires business verification. Zapier/Make.com charge $20-100/month. Most lack support for group messaging to personal WhatsApp groups.

Custom API bots (Telegram Bot API, Slack Webhooks): Free for some platforms but WhatsApp lacks a free group messaging API. Each platform requires separate development.

Manual forwarding: Human copies and pastes. Inconsistent formatting, no deduplication, high effort, doesn't scale.

Open-source WhatsApp bots (WhatsApp-Selenium, PyWhatsapp, whatsappy): Simple message-sending wrappers. Lack email integration, deduplication, configuration management, and production-grade reliability.

1.3 Prior Art Analysis

We conducted a systematic prior art search across Google Patents, Google Scholar, and open-source repositories:

Area	Prior Art	Our Differentiation
Email-to-WhatsApp forwarding	Commercial tools using WhatsApp Business API (n8n, Zapier, SMSEagle)	Browser automation — no API costs, works with personal WhatsApp
Containerized browser automation	Open-source (docker-selenium, docker-chromium-xvfb) for CI/CD testing	Production messaging automation with persistent sessions
ChromeDriver BMP bypass	Stack Overflow workarounds for generic inputs	Adapted for WhatsApp Web's React-based contenteditable inputs
Spreadsheet as config	Patents US8276150B2, US9575950B2 for IT infrastructure management	Applied to messaging pipeline routing — different domain
Pipeline deduplication	Email dedup patents exist broadly	**No prior art found** for dedup at email-to-messaging pipeline boundary
Job posting forwarding	No patents or papers found	**Novel end-to-end pipeline** combining all components

Key finding: While individual techniques exist, the integrated pipeline architecture — combining email ingestion, multi-signal deduplication, spreadsheet-driven configuration, and containerized browser-based delivery — is not covered by any existing patent, paper, or open-source project.

1.4 Contributions

This paper makes the following contributions:

1. **FlowBridge** — a generalized framework for email-to-messaging automation applicable across domains
2. **DedupGuard** — multi-signal deduplication engine operating at the pipeline boundary, combining sender fingerprinting, content similarity (difflib), and identifier matching
3. **ClipCast** — clipboard-based Unicode injection bypassing WebDriver BMP limitations via synthetic ClipboardEvent dispatch
4. **SessionHeal** — self-healing browser session management for containerized environments
5. **SheetConfig** — spreadsheet-as-configuration pattern for non-technical pipeline management
6. **BatchForge** — combined batch messaging reducing recipient-side notification fatigue
7. **Zero-cost architecture** using only free-tier services and open-source tools
8. **Production evaluation** demonstrating 100% delivery accuracy

2. FlowBridge Architecture

2.1 The FlowBridge Pipeline

FlowBridge implements a five-stage pipeline generalizable to any email-to-messaging use case:

```
Stage 1          Stage 2          Stage 3           Stage 4          Stage 5
Email        ->  Content      ->  DedupGuard    ->  BatchForge   ->  ClipCast
Ingestion        Extraction       (Dedup Engine)    (Formatter)      (Delivery)

Gmail/IMAP       Regex/LLM        Fingerprint       Template         Browser
Outlook          HTML parse        Body similarity   Emoji inject     Automation
Exchange         Metadata          ID matching       Batch combine    SessionHeal
                                                     Disclaimer       (Recovery)
```

2.2 SheetConfig — Configuration Layer

```
Google Sheet (SheetConfig – Spreadsheet-as-Configuration)
  |
  +-- Queue Tab: Message pipeline data with status tracking
  |     Columns: ID, Status, Subject, Body, Metadata, Source, Timestamps, RetryCount
  |
  +-- Groups Tab: Target messaging groups (dynamic, editable at runtime)
  |     Column: GroupName
  |
  +-- [Optional] Rules Tab: Filtering/routing rules
        Columns: Keyword, TargetGroup, Priority
```

2.3 SessionHeal — Containerized Execution Environment

```
Docker Container
  |-- Xvfb (X Virtual Framebuffer) :99
  |     |-- Web Browser (renders messaging platform)
  |     |-- x11vnc (optional remote viewing, port 5900)
  |-- Python Agent (Selenium WebDriver)
  |-- Volume Mounts:
        |-- /browser-profile (persistent session)
        |-- /config (credentials, tokens)
```

3. Novel Techniques

3.1 DedupGuard — Multi-Signal Deduplication Engine

Prior work on email deduplication focuses on inbox-level filtering (spam detection, conversation threading). Our contribution is deduplication at the **pipeline boundary** — between email ingestion and message dispatch — using multiple complementary signals:

Signal 1: Sender Fingerprint

```
fingerprint = normalize(subject) + "||" + extract_domain(sender_email)
```

Catches: Same recruiter/sender forwarding the same content. Handles cases where the same role is sent by different people at the same company (shared domain).

Signal 2: Unique Identifier Match Catches: Same record re-queued (accidental re-ingestion, system restarts).

Signal 3: Content Similarity (Historical)

```
similarity = SequenceMatcher(None, body1[:500], body2[:500]).ratio()
if similarity >= 0.70:  # threshold
    mark_duplicate()
```

Catches: Same content with different subject lines, slightly reworded descriptions, formatting differences. Uses first 500 characters for O(n) performance.

Signal 4: Content Similarity (Batch) Same as Signal 3 but applied within the current processing batch to prevent intra-batch duplicates.

Evaluation of dedup accuracy:

Scenario	Signal	Detection
Same subject, same sender	Fingerprint	100%
Same subject, different sender (same org)	Fingerprint	100%
Same subject, different org	Fingerprint	Passes through (correct)
Different subject, same body	Body similarity	93% match → caught
Different subject, different body	All signals	Passes through (correct)
Same ID re-queued	ID match	100%

3.2 ClipCast — Clipboard-Based Unicode Injection

Problem: Selenium's ChromeDriver `send_keys()` only supports Basic Multilingual Plane characters (U+0000–U+FFFF). Emoji characters (U+1F4CB clipboard, U+1F3E2 building, etc.) used in formatted messages cause `ChromeDriver only supports characters in the BMP` errors.

Prior approaches: (a) Avoid emoji entirely, (b) Use pyperclip + Ctrl+V (requires system clipboard access, fails in containers), (c) Use ActionChains (same BMP limitation).

Our approach: Construct and dispatch a synthetic ClipboardEvent via JavaScript:

```
const dt = new DataTransfer();
dt.setData('text/plain', fullUnicodeMessage);
const event = new ClipboardEvent('paste', {
    clipboardData: dt,
    bubbles: true,
    cancelable: true,
});
targetElement.dispatchEvent(event);
```

This bypasses ChromeDriver entirely — the browser's native event handling processes the full Unicode range. The technique works with any contenteditable element including WhatsApp Web's React-rendered input components.

Contribution: While generic ClipboardEvent dispatch is known, applying it specifically to bypass ChromeDriver BMP limitations in production messaging automation — particularly for React-based messaging platforms that handle paste events through custom handlers — is a novel application.

3.3 SessionHeal — Self-Healing Browser Session Management

Containerized browser instances face a unique reliability challenge: Chrome creates lock files (SingletonLock, SingletonCookie, SingletonSocket) in the user data directory. When Chrome crashes inside a container (OOM, network timeout, Xvfb failure), these lock files persist and prevent subsequent launches.

Our approach: Pre-launch cleanup routine:

```
def _cleanup_profile_locks():
    for name in ("SingletonLock", "SingletonCookie", "SingletonSocket"):
        os.remove(os.path.join(profile_path, name))
    subprocess.run(["kill_orphaned_drivers"])
    time.sleep(1)  # filesystem sync
```

Combined with Docker's `restart: unless-stopped` policy, this creates a fully self-healing system:

```
Chrome crash → Container restart → Lock cleanup → Fresh Chrome launch → Session
restored from volume
```

3.4 SheetConfig — Spreadsheet-as-Configuration Pattern

Traditional automation systems store configuration in files (YAML, JSON, .env) or databases, requiring developer access to modify. We introduce a pattern where **the same spreadsheet that stores pipeline data also stores configuration**:

Benefits: - Non-technical users can manage the system (add groups, modify rules) - Built-in audit trail (Google Sheets revision history) - No deployment required for configuration changes - Configuration is read fresh each poll cycle — changes take effect automatically - Collaborative — multiple users can manage simultaneously

Prior art differentiation: Patents US8276150B2 and US9575950B2 cover spreadsheet-based management for IT infrastructure (server provisioning, network configuration). Our application to **messaging pipeline routing** — defining which groups receive which content categories — is a distinct domain not covered by existing patents.

3.5 BatchForge — Combined Batch Messaging

Instead of sending N messages for N items, the framework aggregates all pending items into a single formatted message with headers and separators:

```
*N Alert(s) - Date*

[Item 1 formatted]
======================================
[Item 2 formatted]
======================================
[Item N formatted]
```

This reduces group interactions from 2N (N items × G groups) to G (1 message × G groups), significantly reducing notification fatigue for group members.

4. Example Application: Job Posting Forwarding

We demonstrate the framework through a production deployment forwarding job postings from Gmail to WhatsApp groups.

4.1 Domain-Specific Configuration

Framework Component	Job Forwarding Implementation
Email source	Gmail inbox (recruiter emails, LinkedIn InMail, job boards)
Content extraction	Subject, body, company name, recruiter contact, location
Dedup signals	Subject + recruiter domain, job description similarity
Message format	Emoji-rich structured alert with role, company, requirements, contact
Disclaimer	"Contact the recruiter directly, not me"
Groups	Professional networking WhatsApp groups

4.2 Message Template

```
[clipboard] *Job Alert*
[building] *Company:* {company}
[pin] *{subject}*

{body with requirements and contact info}

[email] Source: {recruiter_email}
[calendar] {date_received}

[warning] Disclaimer: If interested, please reach out
to the contact provided above. Do not contact me
regarding this posting.
```

4.3 Other Potential Applications

The same framework can be deployed for:

- **Sales lead alerts**: CRM notification emails → Slack sales channels
- **Property listings**: MLS alert emails → WhatsApp buyer groups
- **Research paper alerts**: Google Scholar notifications → Lab Telegram groups
- **Price drop alerts**: E-commerce notifications → Customer WhatsApp groups
- **Security alerts**: SIEM notification emails → SOC Discord channels
- **News digests**: Newsletter emails → Topic-specific groups

5. AI-Powered vs Rule-Based: A Comparative Analysis

A critical architectural decision in building email-to-messaging pipelines is whether to employ AI/ML techniques or rule-based processing.

5.1 Approach Overview

Approach	Description	Cost
A: Rule-Based (implemented)	Regex, string matching, keyword filters, difflib similarity	$0/month
B: Local LLM (Ollama)	Locally-hosted open-source LLMs for parsing & summarization	$0/month (8GB+ RAM)
C: Cloud LLM API	Commercial APIs (GPT-4, Claude, Gemini)	$5-50/month

5.2 Detailed Comparison

Dimension	Rule-Based	Local LLM (Ollama)	Cloud LLM API
Monthly cost	$0	$0	$5-50
Hardware requirement	Any machine	8GB+ RAM	Any machine
Processing time/email	< 100ms	5-30s (CPU)	1-3s
Company extraction accuracy	85%	95%	98%
Contact extraction accuracy	90%	95%	98%
Deduplication accuracy	95%	98%	99%
Summarization	N/A (full forward)	85% quality	95% quality
Setup complexity	Easy	Medium	Easy
Offline capability	Partial	Full	No
Deterministic output	Yes	No	No
Maintenance	Low	Medium	Low
Privacy	Data stays local	Data stays local	Data sent to API

5.3 Decision Framework: When to Choose Each

Choose Rule-Based When:

- Zero cost is mandatory
- Volume is low (< 50 emails/day)
- Content is semi-structured (recruiter emails follow patterns)
- Deterministic behavior required
- Minimal infrastructure available
- Privacy is critical

Choose Local LLM When:

- Zero cost required but emails are unstructured
- Privacy is critical (no external API calls)
- Semantic understanding needed (near-duplicate detection)
- Summarization of long content required
- Machine has 8GB+ RAM available

Choose Cloud LLM API When:

- Highest accuracy needed for complex/multi-language content
- Budget of $5-50/month is acceptable
- Low latency required (faster than local CPU inference)
- Enterprise features needed (audit logs, SLA)

5.4 Recommended Hybrid Architecture (For Scale)

For deployments handling >100 emails/day:

```
Email → Rule-Based Parser (fast, free, 85% accuracy)
              |
         Parsed? ──Yes──> Queue
              |
             No
              v
         Local LLM (Ollama, zero cost)
              |
         Parsed? ──Yes──> Queue
              |
             No
              v
         Cloud API (fallback, ~$1/month for failures only)
              |
              v
         Queue
```

This achieves >99% parsing accuracy while keeping costs under $1/month.

6. Evaluation

6.1 Production Deployment

The system was deployed monitoring a Gmail inbox receiving 10-20 job emails daily. Messages were forwarded to two WhatsApp groups with combined membership of 500+ members.

6.2 Results

Metric	Value
Total jobs processed	11 (initial deployment period)
Delivery success rate	100%
Duplicate prevention accuracy	100%
False positive dedup rate	0%
Average send time per group	~20 seconds
Container memory usage	< 500MB
Self-healing recoveries	3 (Chrome crash → auto-recovery)
Container restart survivability	5/5 (session preserved)
Infrastructure cost	**$0/month**

6.3 Deduplication Effectiveness

Duplicate Type	Count	Detected	Method
Same subject, same sender	2	2 (100%)	Fingerprint
Same role, different subject	1	1 (100%)	Body similarity (93%)
Same role, different sender	0	N/A	Would use body similarity
False positive (unique marked dup)	0	N/A	—

7. Limitations and Future Work

7.1 Current Limitations

1. **Platform dependency**: WhatsApp Web UI changes can break selectors (mitigated by multi-selector fallback strategy)
2. **QR code requirement**: Initial setup requires one-time manual QR scan via VNC
3. **Rate limiting**: WhatsApp may throttle automated message sending at high volumes
4. **Single-instance**: Cannot horizontally scale due to WhatsApp session singleton constraint

7.2 Future Work

1. **Multi-platform delivery**: Extend to Telegram (Bot API), Slack (Webhooks), Discord (Webhooks) — each with platform-specific delivery engine
2. **Semantic deduplication**: Replace difflib with sentence-transformer embeddings for higher accuracy
3. **Feedback loop**: Track message engagement (replies, reactions) to improve content selection
4. **MCP integration**: Use Model Context Protocol for direct email access without Apps Script
5. **Rule engine**: Configurable routing rules (keywords → specific groups)
6. **Scheduled digests**: Daily/weekly summary mode instead of real-time forwarding

8. Conclusion

We presented a generalized, zero-cost framework for automated email-to-messaging pipelines. The framework introduces novel techniques in multi-signal deduplication at the pipeline boundary, clipboard-based Unicode injection for browser automation, self-healing containerized session management, and spreadsheet-as-configuration for non-technical pipeline management. Demonstrated through a job posting forwarding deployment, the framework achieves 100% delivery accuracy at zero infrastructure cost. Prior art analysis confirms the integrated pipeline architecture represents a novel contribution not covered by existing patents or publications. The framework is domain-agnostic and applicable to any use case requiring automated email content delivery to messaging platform groups.

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